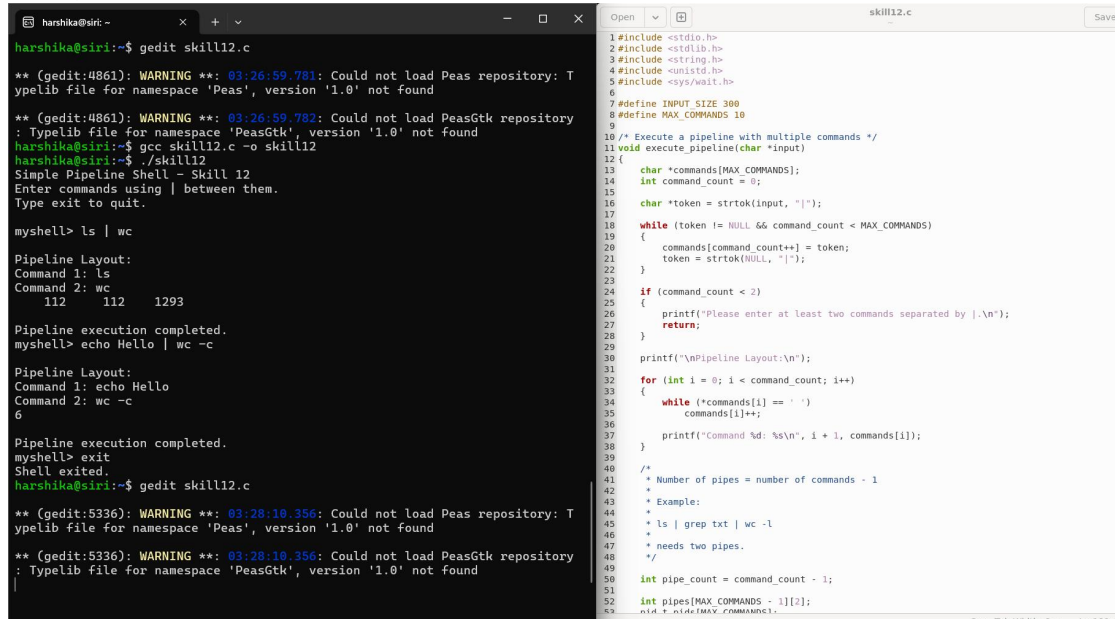


Skill - 12

1. Pipes



The image shows a terminal window on the left and a code editor on the right, illustrating the implementation of a simple pipeline shell.

Terminal Window:

```
harshika@siri:~$ gedit skill12.c
** (gedit:4861): WARNING **: 03:26:59.781: Could not load Peas repository: TypeLib file for namespace 'Peas', version '1.0' not found
** (gedit:4861): WARNING **: 03:26:59.782: Could not load PeasGtk repository: TypeLib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc skill12.c -o skill12
harshika@siri:~$ ./skill12
Simple Pipeline Shell - Skill 12
Enter commands using | between them.
Type exit to quit.

myshell> ls | wc

Pipeline Layout:
Command 1: ls
Command 2: wc
      112      112    1293

Pipeline execution completed.
myshell> echo Hello | wc -c

Pipeline Layout:
Command 1: echo Hello
Command 2: wc -c
6

Pipeline execution completed.
myshell> exit
Shell exited.
harshika@siri:~$ gedit skill12.c
** (gedit:5336): WARNING **: 03:28:10.356: Could not load Peas repository: TypeLib file for namespace 'Peas', version '1.0' not found
** (gedit:5336): WARNING **: 03:28:10.356: Could not load PeasGtk repository: TypeLib file for namespace 'PeasGtk', version '1.0' not found
```

Code Editor (skill12.c):

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <sys/wait.h>
6
7 #define INPUT_SIZE 380
8 #define MAX_COMMANDS 10
9
10 /* Execute a pipeline with multiple commands */
11 void execute_pipeline(char *input)
12 {
13     char *commands[MAX_COMMANDS];
14     int command_count = 0;
15
16     char *token = strtok(input, "|");
17
18     while (token != NULL && command_count < MAX_COMMANDS)
19     {
20         commands[command_count++] = token;
21         token = strtok(NULL, "|");
22     }
23
24     if (command_count < 2)
25     {
26         printf("Please enter at least two commands separated by |.\n");
27         return;
28     }
29
30     printf("\nPipeline Layout:\n");
31
32     for (int i = 0; i < command_count; i++)
33     {
34         while (*commands[i] == ' ')
35             commands[i]++;
36
37         printf("Command %d: %s\n", i + 1, commands[i]);
38     }
39
40     /*
41      * Number of pipes = number of commands - 1
42      *
43      * Example:
44      * ls | grep txt | wc -l
45      * needs two pipes.
46      */
47
48     int pipe_count = command_count - 1;
49
50     int pipes[MAX_COMMANDS - 1][2];
51     pid_t pids[MAX_COMMANDS];
```